ROBOT PROFILE

AGSG Self-Driving Vehicle

WRO Future Engineers · Season 2026 · Portugal | github.com/ImSimao/FE-AGSG-S2026

Team **AGSG** — Joana Silva & Simão Freire · Coach: Tiago Severino · 3 programmable boards, 4 wheels, one rear driven axle

01 AT A GLANCE

FOOTPRINT
200 × 61 × 170 mm
L × W × H; low nose earns the 300 mm bay

MASS
610 g
41% of the 1.5 kg limit (rule 11.2)

DRIVETRAIN
N20 1000 rpm · 3:5
Single motor, metal RC differential

STEERING
Ackermann, 1 servo
Savox SH-0255 MG, metal gears

SPEED
45–60 cm/s
Operating target; ~80 cm/s at rated

POWER
3× 18650 · 11.1 V
20 Wh, three rails, 1.5–2 h runtime

02 CONTROLLERS

BOARD	PROCESSOR	ROLE IN THE VEHICLE	FIRMWARE	INTERFACES
Raspberry Pi Pico 2 *main controller*	RP2350 dual-core, 300 MHz	Navigation, control and sensor fusion: challenge state machine, odometry, PID steering, cruise control	MicroPython	UART0 · UART1 I2C1 · PWM · GPIO
Raspberry Pi Pico 1 *sonar slave*	RP2040	Reads the four SRF04 ultrasonic sensors sequentially and streams framed distances to the main board	MicroPython	UART0 → Pico 2 GPIO17
OpenMV H7 R1 *camera*	STM32H7 + colour camera	Detects the red and green traffic-sign pillars and streams one detection line per frame	MicroPython	UART1 115200 → Pico 2 GPIO9

No external computer is used at runtime: **all perception, planning and control run on the vehicle**; the boards exchange framed UART messages so a dropped byte can never desynchronise the receiver.

03 PROGRAMMING LANGUAGES & TOOLCHAIN

LANGUAGE / TOOL	WHERE AND WHAT FOR
MicroPython	Application firmware on all three boards: the 20 source files in src/ (main controller), SonarSlave/ and Camera/
Python 3 real-time subset	_thread (core 1 control loop), machine (PWM, UART, I2C), utime and the vendored bno08x driver
MicroPico · Thonny · mpremote · OpenMV IDE	Flashing the two Picos and the camera, uploading files over USB
Git + GitHub	Version control and reproducibility: 68 commits, one public repository

Only interpreted MicroPython is used — no compiled or vendor-locked code, so a judge can read and re-flash the whole system from the repository.

04 HOW IT DRIVES

Each run is a closed loop. A state machine picks the driving direction from the side sonars, parks out, follows the outer lane for three laps and parks again. Position is fused from the wheel encoder (distance) and the BNO085 (heading) at 100 Hz, and anchored to the real walls by sonar corrections. A PID controller (kp 0.35, ki 0.003, kd 0.5) steers on cross-track error; an incremental cruise controller holds 45–60 cm/s with limits that keep the wheels from slipping. Green pillars are passed on the left, red on the right; sign colours are learned on lap one and reused on laps two and three.

05 KEY ENGINEERING DECISIONS

DECISION TAKEN	ALTERNATIVE REJECTED	WHY
_thread on two cores	*uasyncio cooperative scheduler*	Scheduler jitter degraded the driving loop; two real cores give deterministic timing
Sonars read one at a time	*All four sonars pinged together*	Concurrent pings produced cross-talk — one sensor reading another's echo
BNO085 game rotation vector	*CMPS12 magnetometer compass*	Magnetic fields on the track caused heading jumps of up to 40° in a single lap
Ultrasonic distance sensing	*IR or LiDAR range sensors*	Wall colour and reflectivity are unknown; ultrasonic readings stay consistent across surfaces

06 BUILDING MATERIALS & MECHANICAL CONSTRUCTION

SUBSYSTEM	MATERIAL / PART	ENGINEERING REASON
Chassis & body	3D-printed black frame; electronics on a single perfboard deck	The body rides above the 100 mm walls while the low front nose keeps the 300 mm parking bay
Drive motor	N20 12 V DC, 1000 rpm, integrated quadrature encoder	Hits the ~80 cm/s target with torque reserve while supplying odometry
Reduction	3:5 gear pair between motor and rear axle	Raises wheel torque and keeps the motor inside its efficient range
Differential	Metal RC differential (Wltoys 144001 type) on the rear axle	Rear wheels turn at different speeds in corners; per-side electronic differentials are banned (rule 11.5)
Steering	Savox SH-0255 MG metal-gear servo + Ackermann linkage	Low backlash and survives load — two SG90 servos broke; Ackermann avoids the wheel slip that would corrupt odometry
Bearings	5 × 11 mm OD / 7 mm ID (rear axle + motor), 2 × 8 mm OD / 3 mm ID (front wheels)	Low-friction rotation on the driven axle and accurate steering pivots
Wheels	Four wheels; the rear pair is driven through the differential	Four-wheel rule; the front pair only steers
Fasteners & mounts	Standard hardware kits, laser-cut and 3D-printed brackets (rules 11.18–11.19)	Fast iteration on sensor and board mounts between test runs

07 POWER, SENSING & SIGNAL CHAIN

SIGNAL CHAIN

4× SRF04 → **Pico 1** → UART0 → **Pico 2** OpenMV H7 → UART1 → **Pico 2** BNO085 → I2C → **Pico 2**
Pico 2 → PWM 20 kHz + dir → TB6612FNG → N20 motor **Pico 2** → PWM 50 Hz → steering servo encoder → A/B → **Pico 2**

RAIL	REGULATOR	VOLTAGE	FEEDS
Motor	LM2596 buck	**11 V**	TB6612FNG H-bridge → N20 motor (≈200 mA at load)
Steering	L7805CV #1	**5 V**	Savox SH-0255 MG servo (≈150 mA, 750 mA stall)
Logic & sensors	L7805CV #2	**5 V**	Pico 2, Pico 1, BNO085, 4× SRF04, OpenMV H7 (≈350 mA)

Power budget. Battery 3 × 18650 Li-ion in series (11.1 V nominal, 1800 mAh, 20 Wh). The motor rail is deliberately separated from the logic rail so switching noise cannot reach the encoder. Estimated load ≈ 4.7 W, ≈ 8.1 W drawn from the battery once regulator losses are counted, giving 1.5–2 h of practical running. The two linear regulators waste about half the input power as heat — a buck on the servo rail is the next planned change.

Sensing. Four SRF04 ultrasonic sensors (front, rear, left and right, measured sequentially to avoid cross-talk) for walls and obstacles; a BNO085 IMU using the game rotation vector without a magnetometer, because a compass heading jumped up to 40° per lap near magnetic fields; an OpenMV H7 colour-blob detector for the red and green traffic signs; and a quadrature encoder on the motor for distance.

08 COMPETITION COMPLIANCE

WRO 2026 REQUIREMENT	STATUS	EVIDENCE / NOTE
Footprint ≤ 300 × 200 × 300 mm	**Met**	200 × 61 × 170 mm
Mass ≤ 1.5 kg	**Met**	610 g — about 41% of the limit (rule 11.2)
4 wheels, one driven axle, mechanical differential (11.5)	**Met**	Rear axle with a Wltoys-type metal differential
Ackermann steering, one steering servo	**Met**	Savox SH-0255 MG
No RF / Bluetooth / Wi-Fi during rounds (11.10)	**Met in rounds**	HC-06 is bench telemetry only; unplugged for competition
Single power switch + start button (9.10–9.11)	**Met**	Single power switch; the car waits for the start button before moving
English documentation on GitHub (chapter 7)	**In progress**	README and journal in English; source comments being translated

REPOSITORY *everything needed to rebuild the car*
models/ CAD & printable parts · **schemes/** wiring · **src/** firmware · **other/** extras · **v-photos/** vehicle · **t-photos/** team · **video/** driving runs

Source: AGSG engineering journal, README and vehicle photos, WRO Future Engineers 2026 season. Mass measured on the complete vehicle, battery included.